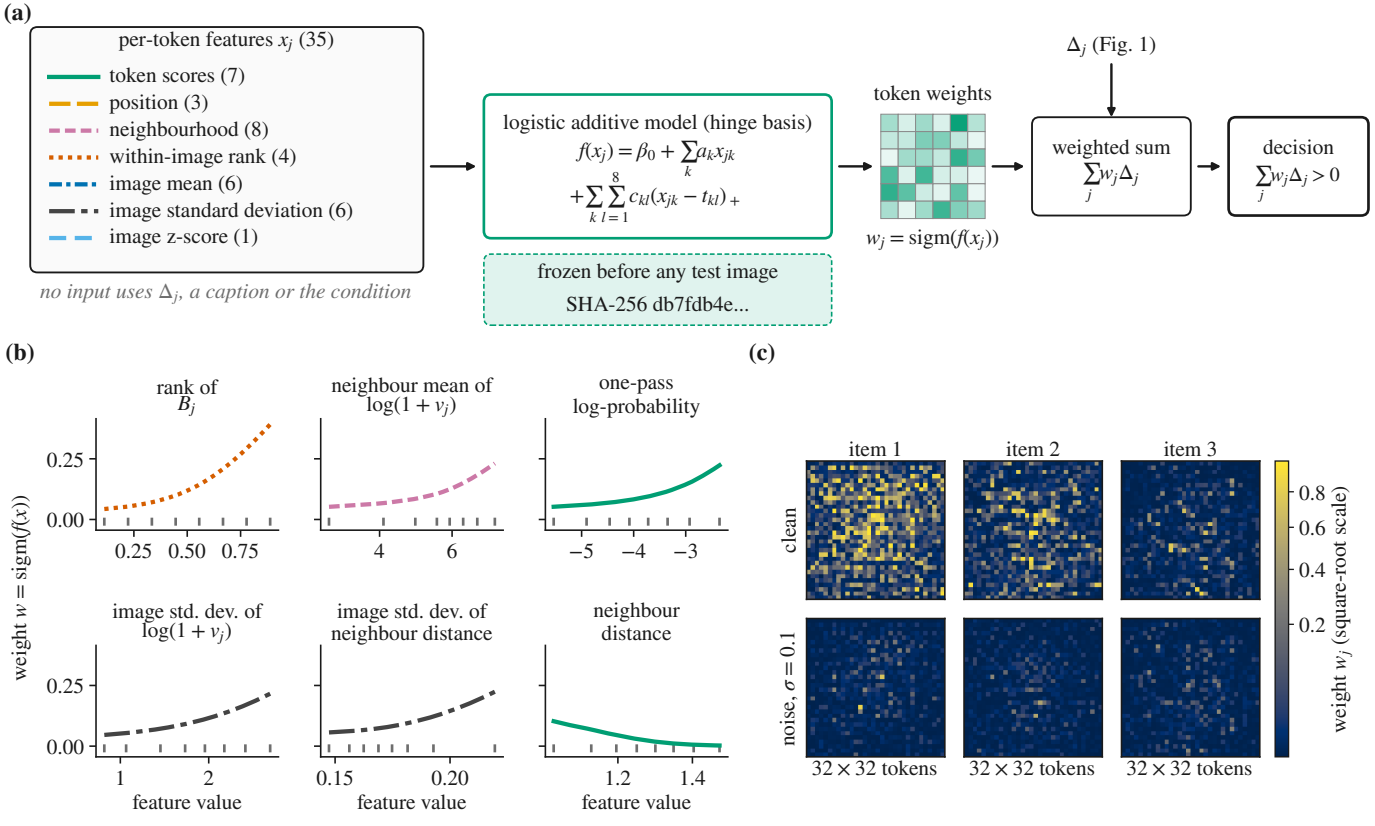
(a)
per-token features x_j (35)
token scores (7)
position (3)
neighbourhood (8)
within-image rank (4)
image mean (6)
image standard deviation (6)
image z-score (1)
no input uses Δ_j, a caption or the condition
logistic additive model (hinge basis)
$f(x_j) = \beta_0 + \sum_k a_k x_{jk} + \sum_k \sum_{l=1}^{8} c_{kl}(x_{jk} - t_{kl})_+$
frozen before any test image
SHA-256 db7fdb4e...
token weights
$w_j = \mathrm{sigm}(f(x_j))$
Δ_j (Fig. 1)
weighted sum $\sum_j w_j \Delta_j$
decision $\sum_j w_j \Delta_j > 0$
(b)
rank of B_j
neighbour mean of $\log(1 + v_j)$
one-pass log-probability
image std. dev. of $\log(1 + v_j)$
image std. dev. of neighbour distance
neighbour distance
weight $w = \mathrm{sigm}(f(x))$
0.25
0.00
0.25 0.50 0.75
4 6
−5 −4 −3
1 2
0.15 0.20
1.2 1.4
feature value
(c)
item 1
item 2
item 3
clean
noise, $\sigma = 0.1$
32 × 32 tokens
0.8
0.6
0.4
0.2
weight w_j (square-root scale)